# Gradient Descent

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**Definition 0.1** (Gradient descent)**.** Let $f : \mathbb{R}^p \to \mathbb{R}$ have first partial derivatives, let $\underset{\sim}{x}^{(0)}$ be a starting point, and let $s^{(0)}, s^{(1)}, \ldots$ be positive **step lengths**. **Gradient descent** [1] computes, for $t = 0, 1, 2, \ldots$,

$$\underset{\sim}{x}^{(t+1)} \stackrel{\text{def}}{=} \underset{\sim}{x}^{(t)} - s^{(t)} \frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{x}^{(t)}).$$

**Example 0.1** (Fixed steps on a parabola)**.** For $f(x) = x^2$, $f'(x) = 2x$, so gradient descent with fixed step length $s$ is

$$\begin{aligned} x^{(t+1)} &= x^{(t)} - 2s\, x^{(t)} \\ &= (1 - 2s)\, x^{(t)}, \end{aligned}$$

and $x^{(t)} = (1 - 2s)^t x^{(0)}$.

- With $s = 0.25$, $x^{(t)} = 0.5^t x^{(0)}$, which tends to the minimizer 0.
- With $s = 0.5$, the first step lands on 0, since $1 - 2s = 0$.
- With $s = 1.5$, $x^{(t)} = (-2)^t x^{(0)}$, which moves away from 0 for any $x^{(0)} \neq 0$: too long a step can make gradient descent fail.

Remark (Gradient descent as Newton's method with a simpler Hessian)

*Remark* 0.1 (Gradient descent as Newton's method with a simpler Hessian). Gradient descent (Definition 0.1) is Newton's method (Definition 5 in Newton's Method) with the Hessian replaced by $\frac{1}{s^{(t)}} \mathbf{I}_p$, so it needs no second derivatives.

**Definition 0.2** (Gradient ascent)**.** Let $f : \mathbb{R}^p \to \mathbb{R}$ have first partial derivatives, let $\underset{\sim}{x}^{(0)}$ be a starting point, and let $s^{(0)}, s^{(1)}, \ldots$ be positive step lengths. **Gradient ascent** [1]

repeats, for $t = 0, 1, 2, \ldots$, the update

$$\underset{\sim}{x}^{(t+1)} \overset{\text{def}}{=} \underset{\sim}{x}^{(t)} + s^{(t)} \frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{x}^{(t)}).$$

**Example 0.2** (Gradient ascent is gradient descent on the negative function)**.** Gradient ascent on $f$ takes the same steps as gradient descent on $-f$, since $\frac{\partial}{\partial \underset{\sim}{x}}(-f) = -\frac{\partial}{\partial \underset{\sim}{x}} f$. In one variable, for $f(x) = -x^2$ the derivative is $f'(x) = -2x$, so gradient ascent with fixed step length $s$ is $x^{(t+1)} = (1 - 2s)\, x^{(t)}$, the iteration of Example 0.1, which tends to the maximizer 0 when $0 < s < 1$.

**Definition 0.3** (Line search)**.** In gradient descent (Definition 0.1), a **line search** [2] is a rule that chooses the step length $s^{(t)}$ by examining the values of $f$ along the line $\left\{\underset{\sim}{x}^{(t)} - s \frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{x}^{(t)}) : s > 0\right\}$.

**Definition 0.4** (Exact line search)**. Exact line search** is the line search (Definition 0.3) that chooses $s^{(t)}$ to minimize $f(\underset{\sim}{x}^{(t)} - s \frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{x}^{(t)}))$ over $s > 0$.

**Example 0.3** (Exact line search on a parabola)**.** For $f(x) = x^2$ and $x^{(t)} = 2$, the derivative is $f'(2) = 4$, so the line is $\{2 - 4s : s > 0\}$ and the values along it are

$$\begin{aligned} \phi(s) &= f(2 - 4s) \quad \text{(the value at step length } s\text{)} \\ &= (2 - 4s)^2 \quad (f(x) = x^2) \end{aligned}$$

$\phi(s) \geq 0$ for every $s$, and

$$\begin{aligned} \phi(0.5) &= (2 - 2)^2 \\ &= 0, \end{aligned}$$

so exact line search chooses $s^{(t)} = 0.5$, which lands on the minimizer $x^{(t+1)} = 0$ in one step.

**Definition 0.5** (Descent direction)**.** Let $f : \mathbb{R}^p \to \mathbb{R}$ have a gradient $\underset{\sim}{g} \overset{\text{def}}{=} \frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{x})$ at $\underset{\sim}{x}$. A direction $\underset{\sim}{d}$ is a **descent direction** for $f$ at $\underset{\sim}{x}$ if $\underset{\sim}{g} \cdot \underset{\sim}{d} < 0$. When $\underset{\sim}{g} \neq \underset{\sim}{0}_{p \times 1}$, $-\underset{\sim}{g}$ is a descent direction, since $\underset{\sim}{g} \cdot (-\underset{\sim}{g}) = -\|\underset{\sim}{g}\|^2 < 0$.

**Example 0.4** (Descent directions for $x_1^2 + x_2^2$ at $(1, 0)^\top$)**.** Let $f(\underset{\sim}{x}) = x_1^2 + x_2^2$ and $\underset{\sim}{x} = (1, 0)^\top$, so

$$\begin{aligned} \underset{\sim}{g} &= (2x_1,\ 2x_2)^\top \\ &= (2, 0)^\top. \end{aligned}$$

- $\underset{\sim}{d} = (-1, 1)^\top$ is a descent direction:

$$\begin{aligned} \underset{\sim}{g} \cdot \underset{\sim}{d} &= -2 + 0 \\ &= -2 \\ &< 0. \end{aligned}$$

- $\underset{\sim}{d} = (0, 1)^\top$ is not: $\underset{\sim}{g} \cdot \underset{\sim}{d} = 0$.
- $-\underset{\sim}{g} = (-2, 0)^\top$ is: $\underset{\sim}{g} \cdot (-\underset{\sim}{g}) = -4 < 0$.

**Theorem 0.1** (A short enough step downhill decreases the function)**.** *Let $f : \mathbb{R}^p \to \mathbb{R}$ have first partial derivatives that are continuous on $\mathbb{R}^p$, and let $\underset{\sim}{x} \in \mathbb{R}^p$ be a point with $\underset{\sim}{g} \overset{def}{=} \frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{x}) \neq \underset{\sim}{0}_{p\times 1}$. Then there is an $\bar{s} > 0$ with $f(\underset{\sim}{x} - s\underset{\sim}{g}) < f(\underset{\sim}{x})$ for every $s \in (0, \bar{s})$: a short enough step in the descent direction $-\underset{\sim}{g}$ (Definition 0.5) decreases $f$.*

Proof

*Proof.* Let $h(s) \overset{\text{def}}{=} f(\underset{\sim}{x} - s\underset{\sim}{g})$. The inner function $s \mapsto \underset{\sim}{x} - s\underset{\sim}{g}$ has derivative $-\underset{\sim}{g}^\top$, so by the vector chain rule

$$\begin{aligned} h'(s) &= -\underset{\sim}{g}^\top \frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{x} - s\underset{\sim}{g}), && \text{(chain rule)} \\ h'(0) &= -\underset{\sim}{g}^\top \frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{x}) && \text{(set } s = 0\text{)} \\ &= -\underset{\sim}{g} \cdot \underset{\sim}{g} && \text{(definition of } \underset{\sim}{g}\text{)} \\ &= -\|\underset{\sim}{g}\|^2 && \text{(squared length)} \\ &< 0. && (\underset{\sim}{g} \neq \underset{\sim}{0}_{p\times 1}) \end{aligned}$$

Since $h'(0) = \lim_{s\to 0}(h(s) - h(0))/s$ is negative, taking $\varepsilon = |h'(0)|$ in the definition of the limit gives an $\bar{s} > 0$ such that $(h(s) - h(0))/s < h'(0) + |h'(0)| = 0$ for $0 < s < \bar{s}$. Multiplying by $s > 0$, $h(s) < h(0)$, that is, $f(\underset{\sim}{x} - s\underset{\sim}{g}) < f(\underset{\sim}{x})$. □

**Example 0.5** (How short is short enough)**.** In Example 0.1, $f(x) = x^2$ at $x \neq 0$ has $g = 2x$, and $f(x - 2sx) = (1 - 2s)^2x^2 < x^2$ exactly when $|1 - 2s| < 1$, that is, when $0 < s < 1$. So Theorem 0.1 holds with $\bar{s} = 1$, and the step $s = 1.5$ of that example was too long.

**Example 0.6** (Gradient descent zigzags)**.** Let $f(\underset{\sim}{x}) = \frac{1}{2}(x_1^2 + b\,x_2^2)$ with $b > 0$, so $\frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{x}) = (x_1,\ bx_2)^\top$ and one gradient step of length $s$ from $\underset{\sim}{x}$ gives $((1-s)x_1,\ (1-bs)x_2)^\top$. For exact line search (Definition 0.4), let $\phi(s) \overset{\text{def}}{=} f$ at that point:

$$\begin{aligned}
\phi(s) &= \tfrac{1}{2}((1-s)^2 x_1^2 + b(1-bs)^2 x_2^2), && \text{(definition of } f\text{)}\\
\phi'(s) &= -(1-s)x_1^2 - b^2(1-bs)x_2^2 && \text{(chain rule, term by term)}\\
&= -x_1^2 + s\,x_1^2 - b^2 x_2^2 + b^3 s\,x_2^2 && \text{(expand the products)}\\
&= s\,(x_1^2 + b^3 x_2^2) - (x_1^2 + b^2 x_2^2). && \text{(collect the terms in } s\text{)}
\end{aligned}$$

For $\underset{\sim}{x} \neq \underset{\sim}{0}_{2\times 1}$, $x_1^2 + b^3 x_2^2 > 0$, so $\phi'$ is increasing in $s$ with a single zero, and $\phi$ is minimized at $s^* = \dfrac{x_1^2 + b^2 x_2^2}{x_1^2 + b^3 x_2^2}$. From $\underset{\sim}{x}^{(0)} = (b, 1)^\top$:

$$\begin{aligned}
s^{(0)} &= \frac{b^2 + b^2}{b^2 + b^3} && \text{(substitute } x_1 = b, x_2 = 1\text{)}\\
&= \frac{2}{1+b}, && \text{(cancel } b^2\text{)}
\end{aligned}$$

and, with $r \overset{\text{def}}{=} \frac{b-1}{b+1}$,

$$\begin{aligned}
1 - s^{(0)} &= \frac{1 + b - 2}{1 + b}\\
&= r, && \text{(common denominator)}\\
1 - b\,s^{(0)} &= \frac{1 + b - 2b}{1 + b}\\
&= -r, && \text{(common denominator)}\\
\underset{\sim}{x}^{(1)} &= (r\,b,\ -r \cdot 1)^\top\\
&= r\,(b, -1)^\top. && \text{(the gradient step from } (b,1)^\top\text{)}
\end{aligned}$$

For $b \neq 1$ the same holds at every step, by induction (when $b = 1$, $r = 0$ and $\underset{\sim}{x}^{(1)}$ is already the minimizer): if $\underset{\sim}{x}^{(t)} = (c\,b,\ \pm c)^\top$ for a number $c \neq 0$, then $s^*$, which depends only on $x_1^2 = c^2 b^2$ and $x_2^2 = c^2$, is again $\frac{2}{1+b}$ (the factor $c^2$ cancels), and the step multiplies the first entry by $r$ and the second by $-r$. So

$$\begin{aligned}
\underset{\sim}{x}^{(t)} &= (b\,r^t,\ (-r)^t)^\top,\\
f(\underset{\sim}{x}^{(t)}) &= \tfrac{1}{2}(b^2 + b)\,r^{2t}\\
&= r^{2t} f(\underset{\sim}{x}^{(0)}).
\end{aligned}$$

For $b \neq 1$, exactly one entry changes sign at every step: the second when $b > 1$ (then $r > 0$), and the first when $b < 1$ (then $r < 0$). So the iterates zigzag toward the minimizer $\underset{\sim}{0}_{2\times 1}$.

- With $b = 3$, $r = \frac{1}{2}$: $\underset{\sim}{x}^{(1)} = (1.5, -0.5)^\top$ and $\underset{\sim}{x}^{(2)} = (0.75, 0.25)^\top$, and $f$ falls from 6 to 1.5 to 0.375, a quarter at each step.
- With $b = 1$, $r = 0$ and one step reaches the minimizer.
- With $b = 0.01$, $r^2 = (0.99/1.01)^2 \approx 0.96$, so $f$ shrinks by only about 4% per step. Newton's method reaches the minimizer of any such $f$ in one step (Theorem 2 in Newton's Method).

**Definition 0.6** (Gradient flow)**.** Let $f : \mathbb{R}^p \to \mathbb{R}$ have first partial derivatives. A **gradient flow** of $f$ is a differentiable curve $\underset{\sim}{x} : [0, \infty) \to \mathbb{R}^p$ that satisfies

$$\frac{d\underset{\sim}{x}(t)}{dt} = -\frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{x}(t)).$$

**Example 0.7** (Gradient descent follows the gradient flow)**.** Let $f(x) = \frac{1}{2}x^2$ on $\mathbb{R}$, so $\frac{\partial}{\partial x} f(x) = x$. The gradient flow (Definition 0.6) from $x(0) = x_0$ solves $\frac{dx(t)}{dt} = -x(t)$, whose solution is

$$x(t) = x_0 \exp\{-t\}.$$

Gradient descent with a fixed step length $s$ multiplies $x$ by $1 - s$ at each step, so after $k$ steps it has reached time $t = ks$ with

$$x^{(k)} = x_0 \, (1 - s)^k.$$

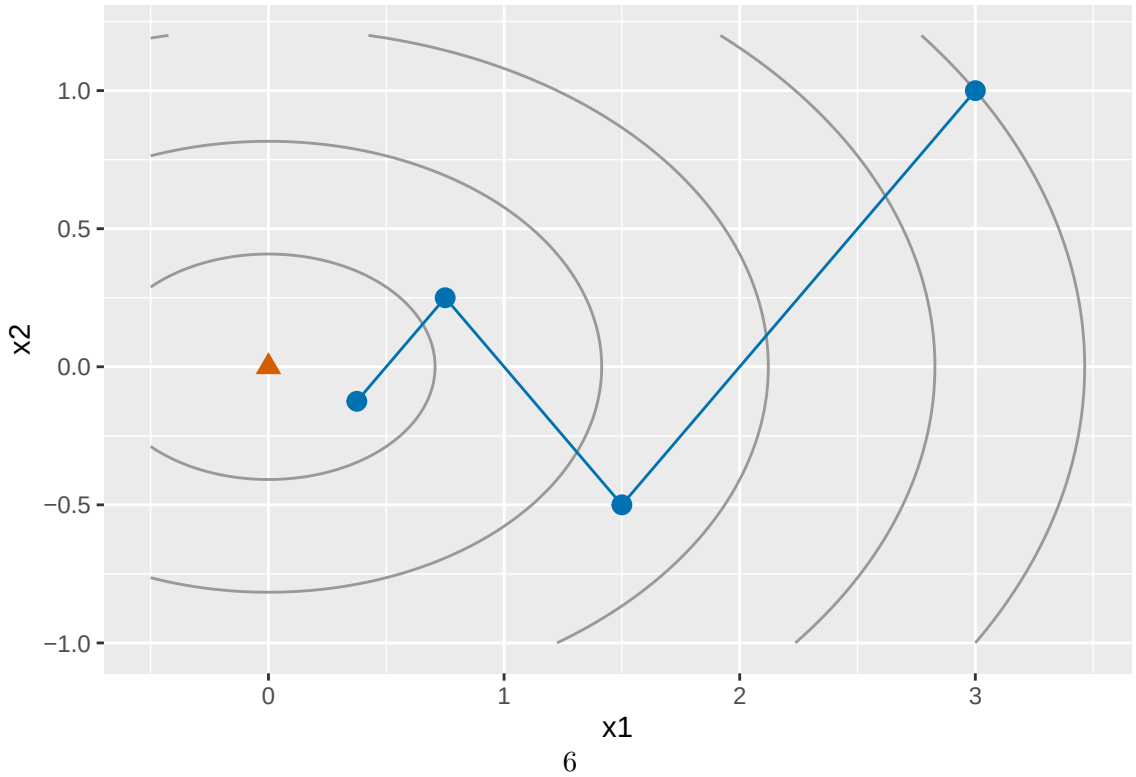
Start at $x_0 = 1$ and run to time $t = 1$:

**R**

```
x0_gf <- 1
t_gf <- 1
s_gf <- c(0.5, 0.01)
descent_gf <- x0_gf * (1 - s_gf)^(t_gf / s_gf)
flow_gf <- x0_gf * exp(-t_gf)
values_gf <- c(descent_gf, flow = flow_gf)
round(values_gf, 4)
#>                 flow
#> 0.2500 0.3660 0.3679
```

**R**

```
gz_x1_gz <- seq(-0.5, 3.5, length.out = 201)
gz_x2_gz <- seq(-1, 1.2, length.out = 201)
gz_grid_gz <- expand.grid(x1 = gz_x1_gz, x2 = gz_x2_gz)
gz_grid_gz$f <- 0.5 * (gz_grid_gz$x1^2 + 3 * gz_grid_gz$x2^2)
gz_steps_gz <- 0:3
gz_path_x1_gz <- 3 * 0.5^gz_steps_gz
gz_path_x2_gz <- c(1, -0.5, 0.25, -0.125)
gz_path_gz <- tibble::tibble(x1 = gz_path_x1_gz, x2 = gz_path_x2_gz)
gz_origin_gz <- tibble::tibble(x1 = 0, x2 = 0)
gz_aes_gz <- ggplot2::aes(x1, x2)
gz_aes_z_gz <- ggplot2::aes(z = f)
gz_breaks_gz <- c(0.25, 1, 2.25, 4, 6)
ggplot2::ggplot(gz_grid_gz, gz_aes_gz) +
  ggplot2::geom_contour(
    gz_aes_z_gz, colour = "grey60", breaks = gz_breaks_gz
  ) +
  ggplot2::geom_path(data = gz_path_gz, colour = "#0072B2") +
  ggplot2::geom_point(data = gz_path_gz, colour = "#0072B2", size = 3) +
  ggplot2::geom_point(
    data = gz_origin_gz, colour = "#D55E00", size = 3, shape = 17
  ) +
  ggplot2::labs(x = "x1", y = "x2")
```






**Python**

```
import matplotlib.pyplot as plt
import numpy as np

gz_x1_gz, gz_x2_gz = np.meshgrid(np.linspace(-0.5, 3.5, 201),
```

**Python**

```
import numpy as np

x0_gf, t_gf = 1.0, 1.0
s_gf = np.array([0.5, 0.01])
descent_gf = x0_gf * (1 - s_gf) ** (t_gf / s_gf)
flow_gf = x0_gf * np.exp(-t_gf)
print(np.round(np.append(descent_gf, flow_gf), 4));
#> [0.25   0.366  0.3679]
```

With step length $s = 0.5$ the descent reaches 0.25, and with $s = 0.01$ it reaches 0.366, against 0.3679 for the flow. The smaller the step length, the closer the descent stays to the flow.

Remark (Gradient descent is Euler's method for the gradient flow)

*Remark* 0.2 (Gradient descent is Euler's method for the gradient flow). Gradient descent with a fixed step length $s$ updates $\underset{\sim}{x}^{(t+1)} = \underset{\sim}{x}^{(t)} - s \frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{x}^{(t)})$. This is Euler's method with time step $s$ for the gradient flow equation.
For a linear function $f$, the velocity of the flow is the negative normal vector.
By the level sets of a linear function, that vector is orthogonal to every level set, so the flow is too.

**Definition 0.7** (Relative change stopping criterion)**.** Let $f$ and $\underset{\sim}{x}^{(0)}, \underset{\sim}{x}^{(1)}, \ldots$ be as in gradient descent, and let $\epsilon > 0$. The **relative change stopping criterion** with tolerance $\epsilon$ stops the iteration at the first $t$ for which $f(\underset{\sim}{x}^{(t)}) \neq 0$ and

$$\frac{\left|f(\underset{\sim}{x}^{(t+1)}) - f(\underset{\sim}{x}^{(t)})\right|}{\left|f(\underset{\sim}{x}^{(t)})\right|} < \epsilon.$$

**Example 0.8** (Stopping a descent on $f(x) = x^2 + 1$)**.** Let $f(x) = x^2 + 1$. Run gradient descent on $f$ from $x^{(0)} = 2$ with step length $s = 0.25$, and stop at tolerance $\epsilon = 0.01$. Each step maps $x$ to $x - 0.25 \cdot 2x = x/2$:

**R**

```r
x_rc <- 2
tol_rc <- 0.01
for (t_rc in 0:100) {
  x_next_rc <- x_rc - 0.25 * 2 * x_rc
  cost_rc <- x_rc^2 + 1
  cost_next_rc <- x_next_rc^2 + 1
  diff_rc <- cost_next_rc - cost_rc
  change_rc <- abs(diff_rc) / cost_rc
  below_tol_rc <- change_rc < tol_rc
  if (below_tol_rc) break
  x_rc <- x_next_rc
}
c(t = t_rc, change = change_rc)
#>           t      change
#> 5.000000000 0.002918288
```

**Python**

```python
f_rc = lambda x: x**2 + 1
x_rc, tol_rc = 2.0, 0.01
for t_rc in range(101):
    x_next_rc = x_rc - 0.25 * 2 * x_rc
    change_rc = abs(f_rc(x_next_rc) - f_rc(x_rc)) / abs(f_rc(x_rc))
    if change_rc < tol_rc:
        break
    x_rc = x_next_rc
print(t_rc, change_rc);
#> 5 0.0029182879377431907
```

The criterion first holds at $t = 5$, where the relative change is 0.00292.

**Definition 0.8** (Rosenbrock function)**.** The **Rosenbrock function** on $\mathbb{R}^2$ is

$$f(\underset{\sim}{x}) \stackrel{\text{def}}{=} 100(x_2 - x_1^2)^2 + (1 - x_1)^2, \qquad \underset{\sim}{x} = (x_1, x_2)^\top.$$

It is a standard test problem for optimizers [3].

**Example 0.9** (Values of the Rosenbrock function)**.** We evaluate the Rosenbrock function at the minimizer $(1, 1)$, at the origin, and at the standard starting point $(-1.2, 1)$. Then we evaluate it along the curve $x_2 = x_1^2$, which is the floor of a narrow, curved valley.

**R**

```
rosenbrock_rf <- function(x1, x2) 100 * (x2 - x1^2)^2 + (1 - x1)^2

points_rf <- list(minimizer = c(1, 1), origin = c(0, 0), start = c(-1.2, 1))
sapply(points_rf, function(p) round(rosenbrock_rf(p[1], p[2]), 3))
#> minimizer    origin     start
#>       0.0       1.0      24.2

# On the curve x2 = x1^2 the first term vanishes, leaving (1 - x1)^2.
x_curve_rf <- c(-1, 0, 0.5, 1, 2)
rbind(
  on_curve = rosenbrock_rf(x_curve_rf, x_curve_rf^2),
  formula = (1 - x_curve_rf)^2
)
#>          [,1] [,2] [,3] [,4] [,5]
#> on_curve    4    1 0.25    0    1
#> formula     4    1 0.25    0    1
```

**Python**

```
import numpy as np

def rosenbrock_rf(x1, x2):
    return 100 * (x2 - x1 ** 2) ** 2 + (1 - x1) ** 2

points_rf = {"minimizer": (1, 1), "origin": (0, 0), "start": (-1.2, 1)}
print({name: round(rosenbrock_rf(*p), 3) for name, p in points_rf.items()})
#> {'minimizer': 0, 'origin': 1, 'start': 24.2}

# On the curve x2 = x1^2 the first term vanishes, leaving (1 - x1)^2.
x_curve_rf = np.array([-1.0, 0.0, 0.5, 1.0, 2.0])
print("on_curve:", rosenbrock_rf(x_curve_rf, x_curve_rf ** 2))
#> on_curve: [4.   1.   0.25 0.   1.  ]
print("formula: ", (1 - x_curve_rf) ** 2)
#> formula:  [4.   1.   0.25 0.   1.  ]
```

The function is 0 at $(1, 1)$ and positive everywhere else, because both of its parts are non-negative and vanish together only when $x_1 = 1$ and $x_2 = 1$. So $(1, 1)$ is the global minimizer. At the starting point the value is 24.2, mostly from the first term, because $(-1.2, 1)$ is off the curve $x_2 = x_1^2$. On the curve the function is $(1 - x_1)^2$, which varies slowly compared with the steep walls.

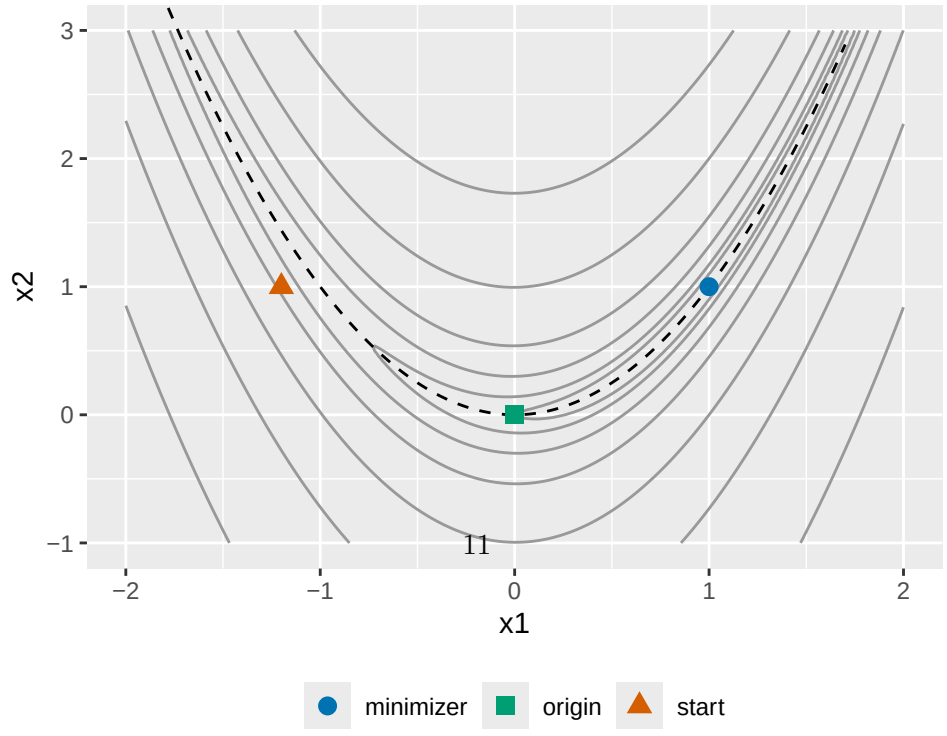
**Definition 0.9** (Basin of attraction)**.** For given step lengths, the **basin of attraction** of a local minimizer $\underset{\sim}{x}^*$ under gradient descent is the set of starting points $\underset{\sim}{x}^{(0)}$ from which the iterates converge to $\underset{\sim}{x}^*$ [4].

**Example 0.10** (Basins of a double-well function)**.** Let $f(x) = x^4 - 2x^2$, so $\frac{\partial}{\partial x} f(x) = 4x(x-1)(x+1)$ and the stationary points are $-1$, 0 and 1. Both $-1$ and 1 are local minimizers, with $f(\pm 1) = -1$, and 0 is a local maximizer. We run gradient descent with the fixed step length $s = 0.05$ for 200 steps from the starting points $-1.5$, $-0.5$, $-0.1$, 0.1, 0.5, 1.5 and 3, and for three steps from $x^{(0)} = 4$.

**R**

**R**

```
library(ggplot2)
rf_grid <- expand.grid(
  x1 = seq(-2, 2, length.out = 201),
  x2 = seq(-1, 3, length.out = 201)
)
rf_grid$f <- rosenbrock_rf(rf_grid$x1, rf_grid$x2)
rf_points <- data.frame(
  name = names(points_rf),
  x1 = sapply(points_rf, `[`, 1),
  x2 = sapply(points_rf, `[`, 2)
)
rf_curve <- data.frame(x1 = seq(-2, 1.7, length.out = 200))
rf_curve$x2 <- rf_curve$x1^2
ggplot(rf_grid, aes(x1, x2)) +
  geom_contour(
    aes(z = f),
    colour = "grey60",
    breaks = c(1, 3, 10, 30, 100, 300, 1000)
  ) +
  geom_line(data = rf_curve, linetype = "dashed") +
  geom_point(aes(colour = name, shape = name), data = rf_points, size = 3) +
  scale_colour_manual(
    values = c(minimizer = "#0072B2", origin = "#009E73", start = "#D55E00")
  ) +
  scale_shape_manual(values = c(minimizer = 16, origin = 15, start = 17)) +
  coord_cartesian(xlim = c(-2, 2), ylim = c(-1, 3)) +
  labs(x = "x1", y = "x2", colour = NULL, shape = NULL) +
  theme(legend.position = "bottom")
```



```r
basin_step_bs <- 0.05
basin_starts_bs <- c(-1.5, -0.5, -0.1, 0.1, 0.5, 1.5, 3)

# 200 steps from every starting point at once.
basin_final_bs <- basin_starts_bs
for (i in 1:200) {
  basin_slope_bs <- 4 * basin_final_bs^3 - 4 * basin_final_bs
  basin_final_bs <- basin_final_bs - basin_step_bs * basin_slope_bs
}
basin_end_bs <- signif(basin_final_bs, 3)
tibble::tibble(start = basin_starts_bs, end = basin_end_bs)
#> # A tibble: 7 x 2
#>   start   end
#>   <dbl> <dbl>
#> 1  -1.5    -1
#> 2  -0.5    -1
#> 3  -0.1    -1
#> 4   0.1     1
#> 5   0.5     1
#> 6   1.5     1
#> 7   3      -1

# The first step from 3.
basin_from_three_bs <- 3 - basin_step_bs * (4 * 3^3 - 4 * 3)

# Three steps from 4: the iterates grow in size.
basin_far_bs <- numeric(4)
basin_far_bs[1] <- 4
for (i in 1:3) {
  basin_slope_bs <- 4 * basin_far_bs[i]^3 - 4 * basin_far_bs[i]
  basin_far_bs[i + 1] <- basin_far_bs[i] - basin_step_bs * basin_slope_bs
}
signif(basin_far_bs, 3)
#> [1]       4.0      -8.0      92.8 -160000.0
```

**Python**

```
basin_step_bs = 0.05

def basin_next_bs(x):
    return x - basin_step_bs * (4 * x ** 3 - 4 * x)

def basin_gd_bs(x0, n_steps=200):
    x = x0
    for _ in range(n_steps):
        x = basin_next_bs(x)
    return x

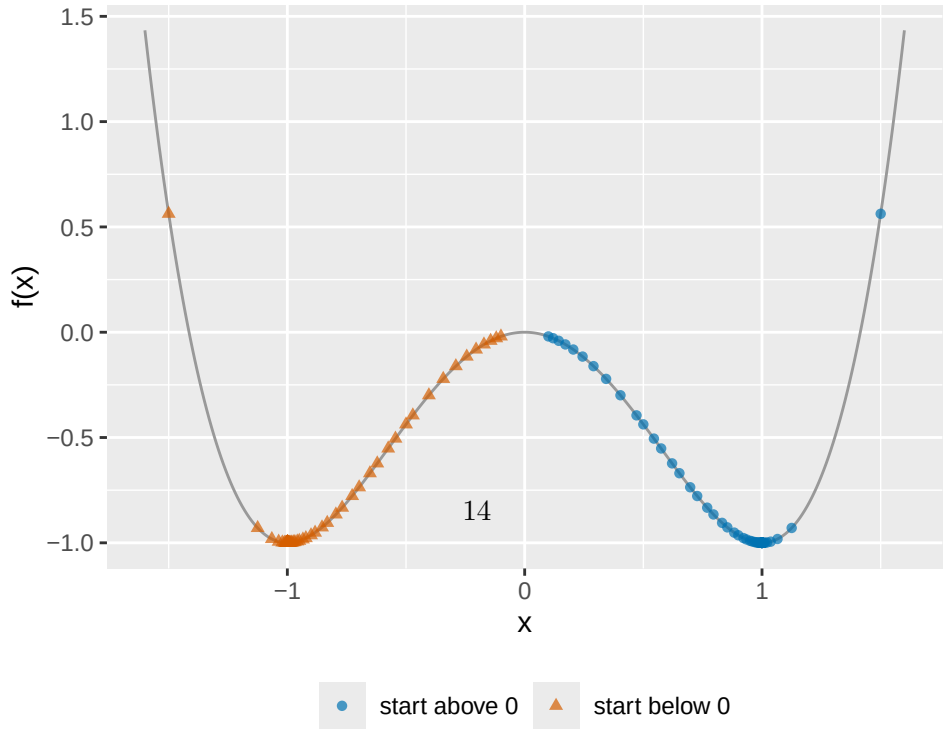
basin_starts_bs = [-1.5, -0.5, -0.1, 0.1, 0.5, 1.5, 3]
for start in basin_starts_bs:
    print(start, float(f"{basin_gd_bs(start):.3g}"))
#> -1.5 -1.0
#> -0.5 -1.0
#> -0.1 -1.0
#> 0.1 1.0
#> 0.5 1.0
#> 1.5 1.0
#> 3 -1.0

# Three steps from 4: the iterates grow in size.
basin_far_bs = [4.0]
for _ in range(3):
    basin_far_bs.append(basin_next_bs(basin_far_bs[-1]))
print([float(f"{v:.3g}") for v in basin_far_bs])
#> [4.0, -8.0, 92.8, -160000.0]
```

Each of the six starts between $-1.5$ and $1.5$ ends at $-1$ or $1$, whichever lies on its own side of 0, so for these starts the sign of the starting point decides which of the two the run reaches. A run from 0 itself never moves, because $\frac{\partial}{\partial x} f(0) = 0$. The basin of attraction of 1 is not all of $(0, \infty)$: the first step from $x^{(0)} = 3$ lands at $-1.8$, on the other side of 0, and the run ends at $-1$. From $x^{(0)} = 4$ the iterates, starting with $x^{(0)}$ itself, are 4, $-8$, 92.8 and $-1.6 \times 10^5$, which grow in absolute value at every step shown. Both of these starts are far from 0, where the derivative is large and so the step $s \frac{\partial}{\partial x} f(x)$ is long. From $x^{(0)} = 3$ that step is $96\,s$ long, so the first step stays on the positive side of 0 only when $s$ is below 0.03125. So the basin of 1 depends on $s$, and with $s = 0.05$ the basin is not a half-line.

**R**

```
# Uses basin_step_bs and basin_starts_bs from the example above.
basin_runs_bf <- matrix(0, nrow = 31, ncol = 6)
basin_runs_bf[1, ] <- basin_starts_bs[1:6]
for (i in 1:30) {
  basin_prev_bf <- basin_runs_bf[i, ]
  basin_slope_bf <- 4 * basin_prev_bf^3 - 4 * basin_prev_bf
  basin_runs_bf[i + 1, ] <- basin_prev_bf - basin_step_bs * basin_slope_bf
}
basin_side_bf <- ifelse(
  basin_starts_bs[1:6] > 0, "start above 0", "start below 0"
)
basin_x_bf <- as.vector(basin_runs_bf)
basin_side_all_bf <- rep(basin_side_bf, each = 31)
basin_paths_bf <- tibble::tibble(x = basin_x_bf, side = basin_side_all_bf)
basin_paths_bf$cost <- basin_paths_bf$x^4 - 2 * basin_paths_bf$x^2
basin_grid_bf <- seq(-1.6, 1.6, length.out = 321)
basin_curve_bf <- tibble::tibble(
  x = basin_grid_bf, cost = basin_grid_bf^4 - 2 * basin_grid_bf^2
)
basin_colours_bf <- c("start above 0" = "#0072B2", "start below 0" = "#D55E00")
ggplot2::ggplot(basin_curve_bf, ggplot2::aes(x, cost)) +
  ggplot2::geom_line(colour = "grey60") +
  ggplot2::geom_point(
    data = basin_paths_bf,
    ggplot2::aes(colour = side, shape = side), alpha = 0.7
  ) +
  ggplot2::scale_colour_manual(values = basin_colours_bf) +
  ggplot2::labs(x = "x", y = "f(x)", colour = NULL, shape = NULL) +
  ggplot2::theme(legend.position = "bottom")
```



**Proposition 0.1** (Divergence of gradient descent on the double-well function)**.** *Let* $f(x) = x^4 - 2x^2$*, and run gradient descent with a fixed step length* $s > 0$*. If* $x^{(0)}$ *satisfies*

$$x^{(0)^2} > 1 + \frac{1}{2s},$$

*then* $\left|x^{(t)}\right| \to \infty$ *as* $t \to \infty$*.*

Proof

*Proof.* The derivative is $\frac{\partial}{\partial x} f(x) = 4x^3 - 4x$, so one step of gradient descent gives

$$\begin{aligned} x^{(t+1)} &= x^{(t)} - s\left(4x^{(t)^3} - 4x^{(t)}\right) && \text{(gradient descent step)} \\ &= -x^{(t)}\left(4sx^{(t)^2} - 4s - 1\right). && \text{(factor out } -x^{(t)}\text{)} \end{aligned}$$

Write $m(x) \stackrel{\text{def}}{=} 4sx^2 - 4s - 1$. Then $\left|x^{(t+1)}\right| = \left|x^{(t)}\right| m(x^{(t)})$ whenever $m(x^{(t)}) \geq 0$. Now $m(x) > 1$ exactly when $x^2 > 1 + 1/(2s)$, and $m(x)$ increases with $x^2$. So if $x^{(t)}$ satisfies the condition, then $\left|x^{(t+1)}\right| > \left|x^{(t)}\right|$, and $x^{(t+1)}$ satisfies it too. By induction on $t$, every iterate satisfies it. Since $\left|x^{(t)}\right|$ increases at every step, $x^{(t)^2} \geq x^{(0)^2}$, and so $m(x^{(t)}) \geq m(x^{(0)}) > 1$. Hence $\left|x^{(t)}\right| \geq m(x^{(0)})^t \left|x^{(0)}\right|$, which tends to $\infty$. □

**Corollary 0.1** (Basins of the double-well function are bounded)**.** *Let* $f(x) = x^4 - 2x^2$*, and run gradient descent with a fixed step length* $s > 0$*. Every start* $x^{(0)}$ *in the basin of attraction of a local minimizer of* $f$ *satisfies*

$$x^{(0)^2} \leq 1 + \frac{1}{2s}.$$

Proof

*Proof.* The iterates from a start in the basin converge to a local minimizer, so they do not diverge to $\infty$. By Proposition 0.1, the start therefore does not satisfy $x^{(0)^2} > 1 + 1/(2s)$. □

**Example 0.11** (Divergent starts on the double-well function for three step lengths)**.** For $f(x) = x^4 - 2x^2$, Proposition 0.1 says that gradient descent diverges to $\infty$ from every start with $\left|x^{(0)}\right|$ above $\sqrt{1 + 1/(2s)}$. This bound is 7.14 for $s = 0.01$, 3.32 for $s = 0.05$, and 2.45 for $s = 0.1$. The derivative $4x^3 - 4x$ is unbounded, as is the second derivative $12x^2 - 4$, so the step $s\frac{\partial}{\partial x} f(x)$ grows without bound with $x$, and a far start overshoots. By Proposition 0.1, every fixed step length fails to converge from some starts. A larger step length lowers the bound, so more starts are guaranteed to diverge. By Corollary 0.1,

each basin of attraction lies within the same bounded interval, which narrows as $s$ grows. In the earlier basin example, $s = 0.05$, and the start 4 lies above the bound 3.32, which is why its iterates grow.

**Definition 0.10** (Plateau)**.** Let $f : \mathbb{R}^p \to \mathbb{R}$ be a function with a gradient at every point of a region, and let $\epsilon > 0$ be a tolerance. The region is a **plateau** of $f$ if the Euclidean norm of the gradient satisfies

$$0 < \left\| \frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{x}) \right\| < \epsilon$$

at every point $\underset{\sim}{x}$ of the region. The function is then nearly flat on the region, and the region contains no stationary point.

**Example 0.12** (A cost with no stationary point)**.**
Let $f(x) = \log(1 + \mathrm{e}^{-x})$. Its derivative is $\frac{\partial}{\partial x} f(x) = -1/(1 + \mathrm{e}^{x})$, which is negative for every $x$, so $f$ has no stationary point at all.
At $x = 0$ the gradient has size 0.5. At $x = 10$ it has size $4.54 \times 10^{-5}$, which is small but not zero. A step of gradient descent subtracts the gradient times the step length. The gradient is negative, so the iterates keep moving to the right, more and more slowly. With tolerance $\epsilon = 10^{-4}$, the region $x \geq 10$ is a plateau (Definition 0.10). A run that enters this plateau has not converged; the run has only slowed down.

**Definition 0.11** (Armijo condition)**.** Let $f : \mathbb{R}^p \to \mathbb{R}$ have first partial derivatives, let $\underset{\sim}{x} \in \mathbb{R}^p$, and let $\underset{\sim}{g} \stackrel{\text{def}}{=} \frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{x})$. A step length $s > 0$ satisfies the **Armijo condition** [5] with constant $c \in (0, 1)$ at $\underset{\sim}{x}$ if

$$f(\underset{\sim}{x} - s\underset{\sim}{g}) \leq f(\underset{\sim}{x}) - c\, s\, \|\underset{\sim}{g}\|^2.$$

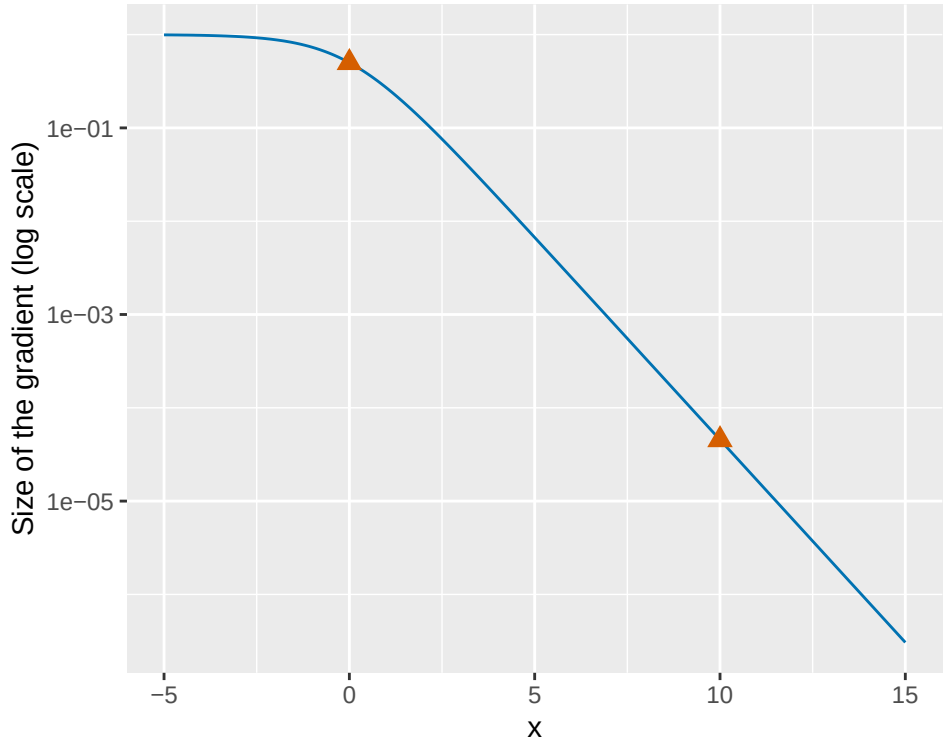
**Example 0.13** (The Armijo condition on a parabola)**.** Let $f(x) = x^2$ and $x = 2$, so $g = f'(2) = 4$ and $\|g\|^2 = 16$. With $c = 1/4$, the condition on a step length $s > 0$ is

$$\begin{aligned} (2 - 4s)^2 &\leq 4 - 4s && (f(2 - 4s) \leq f(2) - c\, s\, \|g\|^2) \\ 4 - 16s + 16s^2 &\leq 4 - 4s && (\text{expand the square}) \\ 16s^2 &\leq 12s && (\text{add } 16s - 4 \text{ to both sides}) \\ s &\leq 3/4. && (\text{divide by } 16s > 0) \end{aligned}$$

So $s = 1/2$ satisfies the condition: $f(0) = 0 \leq 4 - 2 = 2$. And $s = 1$ does not: $f(-2) = 4 > 4 - 4 = 0$.

**R**

```r
plateau_grid_pf <- seq(-5, 15, length.out = 401)
plateau_exp_pf <- exp(plateau_grid_pf)
plateau_size_pf <- 1 / (1 + plateau_exp_pf)
plateau_curve_pf <- tibble::tibble(x = plateau_grid_pf, size = plateau_size_pf)
plateau_marks_x_pf <- c(0, 10)
plateau_marks_exp_pf <- exp(plateau_marks_x_pf)
plateau_marks_size_pf <- 1 / (1 + plateau_marks_exp_pf)
plateau_marks_pf <- tibble::tibble(
  x = plateau_marks_x_pf, size = plateau_marks_size_pf
)
plateau_aes_pf <- ggplot2::aes(x, size)
ggplot2::ggplot(plateau_curve_pf, plateau_aes_pf) +
  ggplot2::geom_line(colour = "#0072B2") +
  ggplot2::geom_point(
    data = plateau_marks_pf, colour = "#D55E00", shape = 17, size = 3
  ) +
  ggplot2::scale_y_log10() +
  ggplot2::labs(x = "x", y = "Size of the gradient (log scale)")
```




**Python**

```python
import matplotlib.pyplot as plt
import numpy as np

plateau_grid_pf = np.linspace(-5, 15, 401)
plateau_marks_pf = np.array([0, 10])
fig, ax = plt.subplots(figsize=(5, 4), layout="constrained")
ax.plot(plateau_grid_pf, 1 / (1 + np.exp(plateau_grid_pf)), color="#0072B2")
#> [<matplotlib.lines.Line2D object at 0x7f8be59f5e50>]
```

**Definition 0.12** (Backtracking line search)**.** Let $s_0 > 0$ be an initial step length, $\beta \in (0,1)$ a shrink factor, and $c \in (0,1)$. **Backtracking line search** [5] is the line search that chooses $s^{(t)}$ to be the largest of $s_0, \beta s_0, \beta^2 s_0, \ldots$ that satisfies the Armijo condition with constant $c$ at $\underset{\sim}{x}^{(t)}$.

**Example 0.14** (Backtracking on a parabola)**.** For $f(x) = x^2$ at $x = 2$ with $c = 1/4$, the Armijo condition holds exactly when $s \le 3/4$ (Example 0.13). With $s_0 = 1$ and $\beta = 1/2$, the trial step lengths are $1, 1/2, 1/4, \ldots$. The first, $s = 1$, exceeds $3/4$ and is rejected. The second, $s = 1/2$, is at most $3/4$ and is accepted. So backtracking chooses $s^{(0)} = 1/2$, after two evaluations of $f$.

**Theorem 0.2** (A bound on the error of gradient descent on a quadratic)**.** *Let $\mathbf{S}$ be a $p \times p$ positive definite matrix with eigenvalues $\lambda_1, \ldots, \lambda_p$, let $\underset{\sim}{c} \in \mathbb{R}^p$, and let $f(\underset{\sim}{x}) = \frac{1}{2}\, \underset{\sim}{x}^\top \mathbf{S} \underset{\sim}{x} - \underset{\sim}{c} \cdot \underset{\sim}{x}$, whose unique minimizer is $\underset{\sim}{x}^* = \mathbf{S}^{-1} \underset{\sim}{c}$ (Theorem 2 in Newton's Method). Gradient descent with a fixed step length $s > 0$ from any $\underset{\sim}{x}^{(0)}$ has errors satisfying, for $t = 1, 2, \ldots$,*

$$\left\| \underset{\sim}{x}^{(t)} - \underset{\sim}{x}^* \right\| \le \rho(s)^t \left\| \underset{\sim}{x}^{(0)} - \underset{\sim}{x}^* \right\|, \qquad \rho(s) \overset{def}{=} \max_{1 \le i \le p} |1 - s\,\lambda_i|.$$

Proof

*Proof.* Let $\underset{\sim}{e}^{(t)} \overset{\text{def}}{=} \underset{\sim}{x}^{(t)} - \underset{\sim}{x}^*$ be the error at step $t$. The gradient of $f$ is $\frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{x}) = \mathbf{S}\underset{\sim}{x} - \underset{\sim}{c}$ (as in the proof of Theorem 2 in Newton's Method), and $\underset{\sim}{c} = \mathbf{S}\underset{\sim}{x}^*$, so

$$\begin{aligned}
\underset{\sim}{e}^{(t+1)} &= \underset{\sim}{x}^{(t)} - s\,(\mathbf{S}\underset{\sim}{x}^{(t)} - \underset{\sim}{c}) - \underset{\sim}{x}^* && \text{(definition of gradient descent)} \\
&= \underset{\sim}{x}^{(t)} - s\,\mathbf{S}(\underset{\sim}{x}^{(t)} - \underset{\sim}{x}^*) - \underset{\sim}{x}^* && (\underset{\sim}{c} = \mathbf{S}\underset{\sim}{x}^*) \\
&= (\mathbf{I}_p - s\,\mathbf{S})\, \underset{\sim}{e}^{(t)}. && \text{(collect the terms in } \underset{\sim}{e}^{(t)})
\end{aligned}$$

By induction on $t$, $\underset{\sim}{e}^{(t)} = (\mathbf{I}_p - s\,\mathbf{S})^t\, \underset{\sim}{e}^{(0)}$. By the spectral theorem, write $\mathbf{S} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^\top$. Then $\mathbf{I}_p - s\,\mathbf{S} = \mathbf{Q}(\mathbf{I}_p - s\,\mathbf{\Lambda})\mathbf{Q}^\top$ because $\mathbf{Q}\mathbf{Q}^\top = \mathbf{I}_p$, and the $t$th power cancels each inner $\mathbf{Q}^\top\mathbf{Q} = \mathbf{I}_p$, so

$$\begin{aligned}
\left\| \underset{\sim}{e}^{(t)} \right\| &= \left\| \mathbf{Q}(\mathbf{I}_p - s\,\mathbf{\Lambda})^t \mathbf{Q}^\top \underset{\sim}{e}^{(0)} \right\| && \text{(the factorization above)} \\
&= \left\| (\mathbf{I}_p - s\,\mathbf{\Lambda})^t \mathbf{Q}^\top \underset{\sim}{e}^{(0)} \right\| && \text{(orthogonal matrices preserve length)} \\
&\le \rho(s)^t \left\| \mathbf{Q}^\top \underset{\sim}{e}^{(0)} \right\| && \text{(each coordinate is multiplied by } (1 - s\lambda_i)^t) \\
&= \rho(s)^t \left\| \underset{\sim}{e}^{(0)} \right\|. && \text{(orthogonal matrices preserve length)}
\end{aligned}$$

The length-preserving steps apply orthogonal matrices preserve length. The inequality holds because $\mathbf{I}_p - s\,\mathbf{\Lambda}$ is diagonal, so its $t$th power multiplies coordinate $i$ by $(1 - s\,\lambda_i)^t$, whose absolute value is at most $\rho(s)^t$. □

**Example 0.15** (The error multiplier for a step length of one quarter)**.** Let $\mathbf{S} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ with eigenvalues 3 and 1, and $\underset{\sim}{c} = \underset{\sim}{0}_{2\times 1}$, so $\underset{\sim}{x}^* = \underset{\sim}{0}_{2\times 1}$. With $s = 0.25$, the factors $1 - s\,\lambda_i$ are 0.25 and 0.75, so $\rho(s) = 0.75$ and

$$\left\|\underset{\sim}{x}^{(t)}\right\| \le 0.75^t \left\|\underset{\sim}{x}^{(0)}\right\|.$$

Along the eigenvector for $\lambda = 3$ the error shrinks fast, and along the eigenvector for $\lambda = 1$ it shrinks slowly; the bound follows the slower one.

**Theorem 0.3** (The best constant step length)**.** *In the setting of Theorem 0.2, let* $\lambda_{\min}$ *and* $\lambda_{\max}$ *be the smallest and largest eigenvalues of* $\mathbf{S}$*. The factor* $\rho(s) = \max_i |1 - s\,\lambda_i|$ *of Theorem 0.2 is smallest over* $s > 0$ *at*

$$s^* = \frac{2}{\lambda_{\min} + \lambda_{\max}},$$

*where*

$$\rho(s^*) = \frac{\kappa - 1}{\kappa + 1}, \qquad \kappa = \kappa(\mathbf{S}),$$

*the condition number of* $\mathbf{S}$*.*

Proof

*Proof.* For $s > 0$, each $1 - s\,\lambda_i$ lies between $u_{\min} \overset{\text{def}}{=} 1 - s\,\lambda_{\max}$ and $u_{\max} \overset{\text{def}}{=} 1 - s\,\lambda_{\min}$, so $|1 - s\,\lambda_i| \le \max(u_{\max}, -u_{\min})$. Both $\lambda_{\min}$ and $\lambda_{\max}$ occur among the $\lambda_i$, and $|u_{\max}| \ge u_{\max}$ and $|u_{\min}| \ge -u_{\min}$, so the maximum is attained:

$$\rho(s) = \max(1 - s\,\lambda_{\min},\ s\,\lambda_{\max} - 1).$$

The first expression is strictly decreasing in $s$, and the second is strictly increasing, and they coincide when

$$\begin{aligned} 1 - s\,\lambda_{\min} &= s\,\lambda_{\max} - 1 && \text{(set the two expressions equal)} \\ 2 &= s\,(\lambda_{\min} + \lambda_{\max}) && \text{(collect the terms in } s\text{)} \\ s &= \frac{2}{\lambda_{\min} + \lambda_{\max}} = s^*. && \text{(divide)} \end{aligned}$$

For $s < s^*$, $\rho(s)$ is at least the decreasing expression, which exceeds its value at $s^*$. For $s > s^*$, $\rho(s)$ is at least the increasing expression, which exceeds its value at $s^*$. So $s^*$ is the minimizer, and

$$\begin{aligned}
\rho(s^*) &= 1 - \frac{2\,\lambda_{\min}}{\lambda_{\min} + \lambda_{\max}} && \text{(first expression at } s^*\text{)} \\
&= \frac{\lambda_{\max} - \lambda_{\min}}{\lambda_{\max} + \lambda_{\min}} && \text{(common denominator)} \\
&= \frac{\kappa - 1}{\kappa + 1}. && \text{(divide the numerator and denominator by } \lambda_{\min}\text{)}
\end{aligned}$$

□

**Example 0.16** (The best step length for eigenvalues 3 and 1)**.** For $\mathbf{S} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ with eigenvalues 3 and 1, Theorem 0.3 gives $s^* = 2/(3+1) = 0.5$, the condition number $\kappa = 3$, and $\rho(s^*) = (3-1)/(3+1) = 0.5$. A grid search over $s = 0.01, 0.02, \ldots, 1$ finds the smallest $\rho(s)$ at $s = 0.5$, the same step length. Each step then multiplies the error by at most 0.5.

**Exercise 0.1** (Which step lengths shrink the error bound)**.** In the setting of Theorem 0.2, find all $s > 0$ with $\rho(s) < 1$, and write the answer as a condition on $s$ that involves $\lambda_{\max}$.

Solution

*Solution* 0.1. $\rho(s) < 1$ says that $|1 - s\,\lambda_i| < 1$ for every $i$, which is $-1 < 1 - s\,\lambda_i < 1$. The right inequality holds for every $s > 0$ because $\lambda_i > 0$. The left inequality is $s\,\lambda_i < 2$, that is, $s < 2/\lambda_i$. These $p$ inequalities, one for each eigenvalue, all hold exactly when the inequality for the largest eigenvalue holds, since it is the most restrictive:

$$0 < s < \frac{2}{\lambda_{\max}}.$$

**Exercise 0.2** (The best step length for eigenvalues 1 and 9)**.** In gradient descent, a fixed step length that is too short converges slowly, and one that is too long diverges. Theorem 0.3 gives the best one from the eigenvalues of $\mathbf{S}$.
Let $\mathbf{S} = \begin{bmatrix} 1 & 0 \\ 0 & 9 \end{bmatrix}$. Using Theorem 0.3, find $s^*$.

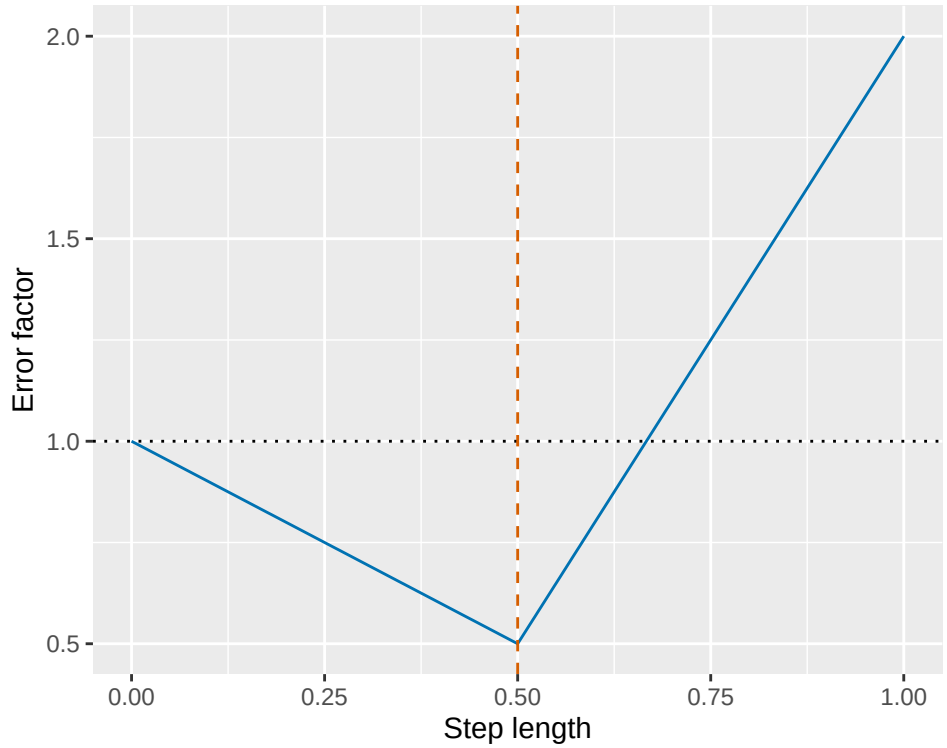
Solution

*Solution* 0.2. $\mathbf{S}$ is diagonal, so its eigenvalues are its diagonal entries 1 and 9. Then

$$s^* = \frac{2}{1+9} = 0.2.$$

**R**

```
bcs_grid_pf <- seq(0, 1, length.out = 401)
bcs_first_pf <- abs(1 - 3 * bcs_grid_pf)
bcs_second_pf <- abs(1 - bcs_grid_pf)
bcs_rho_pf <- pmax(bcs_first_pf, bcs_second_pf)
bcs_curve_pf <- tibble::tibble(s = bcs_grid_pf, rho = bcs_rho_pf)
bcs_aes_pf <- ggplot2::aes(s, rho)
ggplot2::ggplot(bcs_curve_pf, bcs_aes_pf) +
  ggplot2::geom_line(colour = "#0072B2") +
  ggplot2::geom_hline(yintercept = 1, linetype = "dotted") +
  ggplot2::geom_vline(
    xintercept = 0.5, colour = "#D55E00", linetype = "dashed"
  ) +
  ggplot2::labs(x = "Step length", y = "Error factor")
```




**Python**

```
import matplotlib.pyplot as plt
import numpy as np

bcs_grid_pf = np.linspace(0, 1, 401)
bcs_rho_pf = np.maximum(np.abs(1 - 3 * bcs_grid_pf), np.abs(1 - bcs_grid_pf))
fig, ax = plt.subplots(figsize=(5, 4), layout="constrained")
ax.plot(bcs_grid_pf, bcs_rho_pf, color="#0072B2")
#> [<matplotlib.lines.Line2D object at 0x7f8be595a990>]
ax.axhline(1, color="black", linestyle=":")
#> <matplotlib.lines.Line2D object at 0x7f8be595aad0>
ax.axvline(0.5, color="#D55E00", linestyle="--")
#> <matplotlib.lines.Line2D object at 0x7f8be595a210>
ax.set_xlabel("Step length")
```

**Exercise 0.3** (The best error factor for eigenvalues 1 and 9)**.** In the setting of Exercise 0.2, find $\rho(s^*)$.

Solution

*Solution* 0.3. The condition number is $\kappa(\mathbf{S}) = 9/1 = 9$, so by Theorem 0.3

$$\rho(s^*) = \frac{9-1}{9+1} = 0.8.$$

**Exercise 0.4** (Steps to cut the error bound by a factor of ten)**.** In the setting of Exercise 0.2 and Exercise 0.3, find the smallest $t = 1, 2, ...$ with $\rho(s^*)^t \leq 0.1$.

Solution

*Solution* 0.4. From Solution 0.3, $\rho(s^*) = 0.8$. Taking logarithms, $\rho(s^*)^t \leq 0.1$ becomes $t \ln \rho(s^*) \leq \ln 0.1$. Here $\ln \rho(s^*) < 0$, so dividing by it flips the inequality: $t \geq \ln 0.1 / \ln \rho(s^*) \approx 10.32$. The smallest such $t$ is 11.

**Definition 0.13** (Momentum coefficient)**.** Let $\underset{\sim}{g}^{(1)}, \underset{\sim}{g}^{(2)}, ...$ be vectors in $\mathbb{R}^p$, and let $\underset{\sim}{v}^{(t)}$ be a running **velocity** built from them. The **momentum coefficient** $\beta \in [0, 1)$ is the factor that multiplies the previous velocity in

$$\underset{\sim}{v}^{(t)} \stackrel{\text{def}}{=} \beta \underset{\sim}{v}^{(t-1)} + \underset{\sim}{g}^{(t)},$$

for $t = 1, 2, ...$, with $\underset{\sim}{v}^{(0)} \stackrel{\text{def}}{=} \underset{\sim}{0}$.

**Example 0.17** (How far back the velocity remembers)**.** Unrolling the recursion, the vector $\underset{\sim}{g}^{(t-k)}$ enters $\underset{\sim}{v}^{(t)}$ with factor $\beta^k$, for $k < t$. These factors add up to at most $\sum_{k=0}^{\infty} \beta^k = 1/(1-\beta)$. As a rule of thumb, the velocity is therefore dominated by the last $1/(1-\beta)$ vectors.

**R**

```r
beta_mc <- c(0, 0.5, 0.9, 0.99)
k_mc <- 10
weight_mc <- beta_mc^k_mc
memory_mc <- 1 / (1 - beta_mc)
weight_rounded_mc <- signif(weight_mc, 2)
# nolint start: undesirable_function_linter.
rbind(beta = beta_mc, weight = weight_rounded_mc, memory = memory_mc)
#>        [,1]    [,2]  [,3]   [,4]
#> beta      0 0.50000  0.90   0.99
#> weight    0 0.00098  0.35   0.90
#> memory    1 2.00000 10.00 100.00
# nolint end
```

**Python**

```python
beta_mc = [0, 0.5, 0.9, 0.99]
k_mc = 10
for b_mc in beta_mc:
    print(b_mc, float(f"{b_mc**k_mc:.2g}"), round(1 / (1 - b_mc)));
#> 0 0.0 1
#> 0.5 0.00098 2
#> 0.9 0.35 10
#> 0.99 0.9 100
```

At $\beta = 0$ the velocity is the current vector alone, and at $\beta = 0.99$ it combines about 100 vectors.

**Definition 0.14** (Momentum (heavy-ball method))**.** Let $f : \mathbb{R}^p \to \mathbb{R}$ have first partial derivatives, let $\underset{\sim}{x}^{(0)}$ be a starting point, let $s > 0$ be a step length, and let $\beta \in [0, 1)$ be a momentum coefficient. **Momentum** (the **heavy-ball method**) updates, for $t = 1, 2, \ldots$, by

$$\underset{\sim}{v}^{(t)} \stackrel{\text{def}}{=} \beta\, \underset{\sim}{v}^{(t-1)} + \frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{x}^{(t-1)})$$
$$\underset{\sim}{x}^{(t)} \stackrel{\text{def}}{=} \underset{\sim}{x}^{(t-1)} - s\, \underset{\sim}{v}^{(t)},$$

with $\underset{\sim}{v}^{(0)} \stackrel{\text{def}}{=} \underset{\sim}{0}$.

**Example 0.18** (Momentum on a one-dimensional quadratic)**.** Let $f(x) = \frac{1}{2}x^2$, so $\frac{\partial}{\partial x} f(x) = x$. Start at $x^{(0)} = 1$ with $s = 0.1$, and take 20 updates with $\beta = 0$, which is

plain gradient descent, and with $\beta = 0.9$:

**R**

```
x0_mo <- 1
s_mo <- 0.1
beta_mo <- c(plain = 0, momentum = 0.9)
n_mo <- 20
x_mo <- rep(x0_mo, 2)
v_mo <- c(0, 0)
path_mo <- matrix(NA_real_, nrow = n_mo, ncol = 2)
for (k_mo in seq_len(n_mo)) {
  v_mo <- beta_mo * v_mo + x_mo
  x_mo <- x_mo - s_mo * v_mo
  path_mo[k_mo, ] <- x_mo
}
last_mo <- path_mo[n_mo, ]
lowest_mo <- apply(path_mo, 2, min)
# nolint start: undesirable_function_linter.
summary_mo <- rbind(last = last_mo, lowest = lowest_mo)
# nolint end
round(summary_mo, 4)
#>          [,1]    [,2]
#> last   0.1216  0.3447
#> lowest 0.1216 -0.6037
```

**Python**

```
s_mo, n_mo = 0.1, 20
for beta_mo in (0.0, 0.9):
    x_mo, v_mo, path_mo = 1.0, 0.0, []
    for _ in range(n_mo):
        v_mo = beta_mo * v_mo + x_mo
        x_mo = x_mo - s_mo * v_mo
        path_mo.append(x_mo)
    print(beta_mo, round(path_mo[-1], 4), round(min(path_mo), 4));
#> 0.0 0.1216 0.1216
#> 0.9 0.3447 -0.6037
```

After 20 updates, plain descent has reached 0.122 and momentum 0.345. Momentum overshoots the minimizer 0, down to -0.604 at update 9, and then swings back. In this

run, with a small step length, the swing leaves momentum behind plain descent after 20 updates.

**Definition 0.15** (Adaptive method)**.** Let $f : \mathbb{R}^p \to \mathbb{R}$ have first partial derivatives, let $\underset{\sim}{x}^{(0)}$ be a starting point, let $s > 0$ be a step length, let $\rho \in [0, 1)$ be a decay factor, and let $\epsilon > 0$. Write $g_j^{(t)}$ for coordinate $j$ of $\frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{x}^{(t-1)})$. An **adaptive method** is defined, for $t = 1, 2, \ldots$ and each coordinate $j = 1, \ldots, p$,

$$u_j^{(t)} \stackrel{\text{def}}{=} \rho\, u_j^{(t-1)} + (1 - \rho) \left(g_j^{(t)}\right)^2$$

$$x_j^{(t)} \stackrel{\text{def}}{=} x_j^{(t-1)} - s\, \frac{g_j^{(t)}}{\sqrt{u_j^{(t)}} + \epsilon},$$

with $u_j^{(0)} \stackrel{\text{def}}{=} 0$.

**Example 0.19** (Two coordinates on very different scales)**.** Suppose the two coordinates of the gradient start at $g_1^{(1)} = 100$ and $g_2^{(1)} = 0.01$, and both shrink by a factor of 0.5 at each update. Take $\rho = 0.9$, and look at the first 5 updates. Dividing by the root of the running average gives the factor $g_j^{(t)} / \sqrt{u_j^{(t)}}$ that multiplies $s$ in the update of each coordinate:

**R**

```r
s_am <- c(100, 0.01)
rho_am <- 0.9
q_am <- 0.5
n_am <- 5
k_am <- seq_len(n_am) - 1
g_am <- outer(q_am^k_am, s_am)
u_am <- matrix(0, n_am, 2)
u_prev_am <- c(0, 0)
for (t_am in seq_len(n_am)) {
  u_prev_am <- rho_am * u_prev_am + (1 - rho_am) * g_am[t_am, ]^2
  u_am[t_am, ] <- u_prev_am
}
factor_am <- g_am / sqrt(u_am)
round(factor_am, 3)
#>       [,1]  [,2]
#> [1,] 3.162 3.162
#> [2,] 1.474 1.474
#> [3,] 0.755 0.755
#> [4,] 0.395 0.395
#> [5,] 0.208 0.208
```

**Python**

```python
import numpy as np

s_am, rho_am, q_am, n_am = [100, 0.01], 0.9, 0.5, 5
g_am = np.outer(q_am ** np.arange(n_am), s_am)
u_am = np.zeros(2)
factor_am = []
for t_am in range(n_am):
    u_am = rho_am * u_am + (1 - rho_am) * g_am[t_am] ** 2
    factor_am.append(g_am[t_am] / np.sqrt(u_am))
print(np.round(factor_am, 3));
#> [[3.162 3.162]
#>  [1.474 1.474]
#>  [0.755 0.755]
#>  [0.395 0.395]
#>  [0.208 0.208]]
```

Plain gradient descent would move the first coordinate $10^4$ times as far as the second. The factors are the same for both coordinates, so the scale of a coordinate no longer

matters. They also shrink from 3.16 to 0.21, because the running average still holds the earlier, larger values. Dividing by the current gradient alone would give a factor of 1 at every update.

Remark (Early adaptive methods)

*Remark* 0.3 (Early adaptive methods). RMSProp is an early example of an adaptive method [6]. AdaGrad, an earlier method, replaces the exponentially weighted average by a plain sum of squared gradients, $u_j^{(t)} = u_j^{(t-1)} + \left(g_j^{(t)}\right)^2$.

**Definition 0.16** (Adam)**.** Let $f$, $\underset{\sim}{x}^{(0)}$, $s$ and $\epsilon$ be as in an adaptive method, and let $\beta_1, \beta_2 \in [0, 1)$, where $\beta_2$ plays the role of the decay factor $\rho$. Write $\underset{\sim}{g}^{(t)}$ for $\frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{x}^{(t-1)})$. The **Adam** method [7] is defined, for $t = 1, 2, \ldots$,

$$
\begin{aligned}
\underset{\sim}{\mu}^{(t)} &\stackrel{\text{def}}{=} \beta_1 \, \underset{\sim}{\mu}^{(t-1)} + (1 - \beta_1) \, \underset{\sim}{g}^{(t)} \\
u_j^{(t)} &\stackrel{\text{def}}{=} \beta_2 \, u_j^{(t-1)} + (1 - \beta_2) \left(g_j^{(t)}\right)^2 \\
\underset{\sim}{\hat{\mu}}^{(t)} &\stackrel{\text{def}}{=} \frac{\underset{\sim}{\mu}^{(t)}}{1 - \beta_1^t} \\
\hat{u}_j^{(t)} &\stackrel{\text{def}}{=} \frac{u_j^{(t)}}{1 - \beta_2^t} \\
x_j^{(t)} &\stackrel{\text{def}}{=} x_j^{(t-1)} - s \, \frac{\hat{m}_j^{(t)}}{\sqrt{\hat{u}_j^{(t)}} + \epsilon},
\end{aligned}
$$

for each coordinate $j = 1, \ldots, p$, with $\underset{\sim}{\mu}^{(0)} \stackrel{\text{def}}{=} \underset{\sim}{0}$ and $u_j^{(0)} \stackrel{\text{def}}{=} 0$.

**Example 0.20** (The first Adam step ignores the size of the gradient)**.** Let $f(x) = \frac{1}{2}x^2$, so $\frac{\partial}{\partial x} f(x) = x$, and take $s = 0.1$, $\beta_1 = 0.9$, $\beta_2 = 0.999$ and $\epsilon = 10^{-8}$. At $t = 1$ the corrections for the zero start of $\underset{\sim}{\mu}$ and $u_j$ cancel the factors $1 - \beta_1$ and $1 - \beta_2$, so $\hat{m}^{(1)} = g^{(1)}$ and $\hat{u}^{(1)} = \left(g^{(1)}\right)^2$. The first step is then $s \, g^{(1)} / (|g^{(1)}| + \epsilon)$, which is about $s$ in size. Start at $x^{(0)} = 2$ and at $x^{(0)} = 2000$:

**R**

```r
s_ad <- 0.1
beta1_ad <- 0.9
beta2_ad <- 0.999
eps_ad <- 1e-8
x0_ad <- c(2, 2000)
g_ad <- x0_ad
m_ad <- (1 - beta1_ad) * g_ad
u_ad <- (1 - beta2_ad) * g_ad^2
m_hat_ad <- m_ad / (1 - beta1_ad)
u_hat_ad <- u_ad / (1 - beta2_ad)
step_ad <- s_ad * m_hat_ad / (sqrt(u_hat_ad) + eps_ad)
round(step_ad, 4)
#> [1] 0.1 0.1
```

**Python**

```python
import numpy as np

s_ad, beta1_ad, beta2_ad, eps_ad = 0.1, 0.9, 0.999, 1e-8
g_ad = np.array([2.0, 2000.0])
m_hat_ad = (1 - beta1_ad) * g_ad / (1 - beta1_ad)
u_hat_ad = (1 - beta2_ad) * g_ad ** 2 / (1 - beta2_ad)
print(np.round(s_ad * m_hat_ad / (np.sqrt(u_hat_ad) + eps_ad), 4));
#> [0.1 0.1]
```

Both first steps have size 0.1, though the gradients differ by a factor of 1000. Plain gradient descent would move $s\,g^{(1)}$, which is 0.2 and 200.

**Definition 0.17** (Piecewise constant schedule)**.** Let $s^{(0)} > 0$ be the step length before update 1, let $\gamma \in (0,1)$ be a drop factor, and let $t_1 < t_2 < \cdots$ be update counts. The **piecewise constant schedule** is

$$s^{(t)} \overset{\text{def}}{=} s^{(0)}\,\gamma^{k(t)}, \qquad k(t) \overset{\text{def}}{=} \big|\{j : t_j \le t\}\big|.$$

**Example 0.21** (A ten-fold drop twice)**.** Take $s^{(0)} = 0.1$ and $\gamma = 0.1$, with drops at $t_1 = 30$ and $t_2 = 60$. The step length just before and at each drop is:

**R**

```
s0_pc <- 0.1
decay_pc <- 0.1
drops_pc <- c(30, 60)
t_pc <- c(0, 29, 30, 59, 60, 90)
dropped_pc <- outer(drops_pc, t_pc, "<=")
k_pc <- colSums(dropped_pc)
step_pc <- s0_pc * decay_pc^k_pc
tibble::tibble(t = t_pc, step_length = step_pc)
#> # A tibble: 6 x 2
#>       t step_length
#>   <dbl>       <dbl>
#> 1     0       0.1
#> 2    29       0.1
#> 3    30       0.01
#> 4    59       0.01
#> 5    60       0.001
#> 6    90       0.001
```

**Python**

```
s0_pc, decay_pc, drops_pc = 0.1, 0.1, [30, 60]
for t_pc in [0, 29, 30, 59, 60, 90]:
    k_pc = sum(d_pc <= t_pc for d_pc in drops_pc)
    print(t_pc, round(s0_pc * decay_pc ** k_pc, 6));
#> 0 0.1
#> 29 0.1
#> 30 0.01
#> 59 0.01
#> 60 0.001
#> 90 0.001
```

The step length stays fixed between drops. It falls by the factor $\gamma$ exactly at $t = 30$ and $t = 60$.

**Definition 0.18** (Linear decay)**.** Let $\tau$ be a number of updates, and let $s^{(0)}, s^{(\tau)} > 0$ be the step lengths before update 1 and after update $\tau$. The **linear decay** schedule is

$$s^{(t)} \stackrel{\text{def}}{=} (1 - \zeta)\, s^{(0)} + \zeta\, s^{(\tau)}, \qquad \zeta \stackrel{\text{def}}{=} \min\left(\frac{t}{\tau}, 1\right).$$

**Example 0.22** (Linear decay between two step lengths)**.** Take $s^{(0)} = 0.1$, $s^{(\tau)} = 0.001$ and $\tau = 90$. The step length at $t = 0, 45, 90, 120$ is:

**R**

```
s0_ld <- 0.1
s_end_ld <- 0.001
tau_ld <- 90
t_ld <- c(0, 45, 90, 120)
frac_ld <- pmin(t_ld / tau_ld, 1)
step_ld <- (1 - frac_ld) * s0_ld + frac_ld * s_end_ld
tibble::tibble(t = t_ld, step_length = step_ld)
#> # A tibble: 4 x 2
#>       t step_length
#>   <dbl>       <dbl>
#> 1     0      0.1
#> 2    45      0.0505
#> 3    90      0.001
#> 4   120      0.001
```

**Python**

```
s0_ld, sT_ld, tau_ld = 0.1, 0.001, 90
for t_ld in [0, 45, 90, 120]:
    frac_ld = min(t_ld / tau_ld, 1)
    print(t_ld, round((1 - frac_ld) * s0_ld + frac_ld * sT_ld, 6));
#> 0 0.1
#> 45 0.0505
#> 90 0.001
#> 120 0.001
```

The step length falls by the same amount at each update. It then stays at 0.001 for $t \geq \tau$.

**Definition 0.19** (Exponential decay)**.** Let $s^{(0)} > 0$ be the step length before update 1, and let $\gamma \in (0, 1)$ be a decay factor. The **exponential decay** schedule is

$$s^{(t)} \stackrel{\text{def}}{=} s^{(0)} \gamma^t.$$

**Example 0.23** (The same factor over two run lengths)**.** Take $s^{(0)} = 0.1$ and $\gamma = 0.99$. The step length at $t = 1000$ and at $t = 100000$ is:

**R**

```
s0_ed <- 0.1
decay_ed <- 0.99
t_ed <- c(1000, 100000)
step_ed <- s0_ed * decay_ed^t_ed
tibble::tibble(t = t_ed, step_length = step_ed)
#> # A tibble: 2 x 2
#>        t step_length
#>    <dbl>       <dbl>
#> 1   1000  0.00000432
#> 2 100000  0
```

**Python**

```
s0_ed, decay_ed = 0.1, 0.99
for t_ed in [1000, 100000]:
    print(t_ed, s0_ed * decay_ed ** t_ed);
#> 1000 4.317124741065786e-06
#> 100000 0.0
```

At $t = 1000$ the step length is about $4.3 \times 10^{-6}$. At $t = 100000$ it is 0 in floating-point arithmetic. One factor gives two very different schedules.

**Definition 0.20** (Cosine decay)**.** Let $\tau$ be a number of updates, and let $s^{(0)} > 0$ be the step length before update 1. The **cosine decay** schedule [8] is

$$s^{(t)} \stackrel{\text{def}}{=} \frac{s^{(0)}}{2}\left(1 + \cos\frac{\pi t}{\tau}\right), \qquad t = 0, 1, \ldots, \tau.$$

**Example 0.24** (Cosine decay over a fixed run)**.** Take $s^{(0)} = 0.1$ and $\tau = 90$. The step length at $t = 0, 15, 45, 75, 90$ is:

**R**

```r
s0_cd <- 0.1
tau_cd <- 90
t_cd <- c(0, 15, 45, 75, 90)
angle_cd <- pi * t_cd / tau_cd
step_cd <- s0_cd / 2 * (1 + cos(angle_cd))
tibble::tibble(t = t_cd, step_length = step_cd)
#> # A tibble: 5 x 2
#>       t step_length
#>   <dbl>       <dbl>
#> 1     0     0.1
#> 2    15     0.0933
#> 3    45     0.05
#> 4    75     0.00670
#> 5    90     0
```

**Python**

```python
import math

s0_cd, tau_cd = 0.1, 90
for t_cd in [0, 15, 45, 75, 90]:
    print(t_cd, round(s0_cd / 2 * (1 + math.cos(math.pi * t_cd / tau_cd)), 6));
#> 0 0.1
#> 15 0.093301
#> 45 0.05
#> 75 0.006699
#> 90 0.0
```

The step length is 0.1 at the start, half of that at $t = 45$, and zero at $t = 90$. It changes slowly near both ends and fastest in the middle.

Remark (Restarting the cosine schedule)

*Remark* 0.4 (Restarting the cosine schedule). Restarting the schedule periodically is a common variant of cosine decay [8]: jump the step length back to $s^{(0)}$ and decay again. One motivation is that the jump can move the iterates out of one basin of attraction and into a better one.

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